

Analytical architecture of the study

From raw source files to final validation

DATA ASSEMBLY · arithmetic, not inference

Stage 0 Source ingestion

§ 4.2 – 4.6

FC-16 Vol. II Ch. 5 annexures · DPIIT state-wise FDI · MoSPI GSDP (2011-12 series, current prices) · Census 2011 Primary Census Abstract

Stage 1 Panel construction

§ 4.1, 4.3 – 4.5

Balanced 10 × 5 panel · ratios as per cent of GSDP · $\ln(\text{FDI})$ · year-on-year GSDP growth · COVID dummy · 48 usable FDI cells of 50

Stage 2 Composite discipline ranking

§ 4.7, 6.2

Four fiscal ratios ranked across the ten states, then averaged into a composite · weighting sensitivity checked against realised FDI

BRANCH A · classical

Stage 3 Panel estimation

§ 6.3 – 6.4

Models 1 and 2, each estimated twice: fixed effects with Driscoll-Kraay errors, and pooled OLS with Census controls

Stage 4 Diagnostics & inference

§ 6.7

Variance inflation · poolability F · Hausman · Pesaran CD · Wooldridge AR(1) against $\rho = -0.5$ · wild cluster bootstrap, the arbiter at ten clusters

Stage 5 Perturbation

§ 6.5, 7.1

Refits dropping the COVID year, Uttar Pradesh and Jharkhand in turn · year fixed effects against a single dummy · 132-specification curve · Mundlak

BRANCH B · machine learning

Stage 6 Regularised fits

§ 5.2, 6.6

Five time-varying regressors only. LASSO at a state-blocked cross-validated penalty · Elastic Net · forest via SHAP

Stage 7 Leakage audit

§ 6.6, 7.4

Urbanisation diagnosed as a disguised state identifier; branch refitted without time-invariant controls, then rerun with them for comparison

Stage 8 Stability layer

§ 6.6

Ten leave-one-state-out SHAP refits · stability selection, 500 subsamples · state-permuted null · within specification · out-of-sample R^2 three ways

CONVERGENCE

Stage 9 Convergence test — FINAL VALIDATION

§ 5.3, 6.6, 7.1

Five methods over the same 48 observations: pooled panel regression · Spearman rank correlation · LASSO · Elastic Net · SHAP importance. They are not five independent draws and are not treated as one — LASSO and Elastic Net differ only in the penalty, and the rank correlation is significant only with Uttar Pradesh excluded. What the design buys is agreement across three method families: a parametric panel estimator, a regularised linear selector, and a non-parametric tree ensemble. Disagreements are reported.

Stage 10 Residual case analysis

§ 7.3

Jharkhand and Uttar Pradesh examined by name as departures from the fitted pattern. Uttar Pradesh is a censored series and a genuine counterexample both: scaling its observed years still leaves it ninth of ten on realised investment.